

SVAttack: Spatial-Viewpoint Transfer Attack on Graph Convolutional Skeleton Action Recognition

Zhize Wu, Huanxin Zou, Thomas Weise, Xiaofeng Wang, Chenyang Bu, Peiya Li, Teng Li, and Mengyuan Liu

Abstract—Skeleton-based action recognition with graph convolutional networks has achieved remarkable success and is increasingly deployed in safety-critical applications. However, existing adversarial attack methods on skeleton-based models suffer from limited transferability across architectures and often exhibit noticeable perceptual distortion. In this work, we investigate the architectural inductive biases that fundamentally constrain transferability in skeleton-based action recognition. Based on a structural and gradient-level analysis, we propose a spatial gradient damper to suppress model-specific spatial biases, together with a viewpoint-based constraint mechanism to improve perceptual imperceptibility. Extensive experimental results on 3 benchmark datasets, 9 skeleton-based action recognition models, 7 attack methods, and 2 defense methods demonstrate that the proposed approach significantly enhances transferability while maintaining visually plausible skeleton motions, establishing a new state of the art for transferable adversarial attacks on skeleton-based action recognition models. Our code is available at <https://github.com/MilkyWay123123/SVAttack>.

Index Terms—Architectural Inductive Biases, Graph Convolution, Transfer Attack, Action Recognition.

I. INTRODUCTION

Skeleton-based action recognition powered by graph convolutional networks (GCNs) has achieved remarkable progress in recent years [1], [2], [3]. Such methods have been widely deployed in sports training and rehabilitation assessment, public safety monitoring, and medical surveillance [4], [5], [6]. Meanwhile, attacks against AI-powered systems have increasingly attracted public attention and are now recognized as a serious security threat [7]. Given that skeleton-based action recognition systems are often deployed in safety-critical scenarios, understanding their vulnerability to adversarial attacks is of fundamental importance.

Existing adversarial attacks are commonly categorized into white-box, black-box, and transfer attacks according to the attacker's access to the target model. White-box attacks assume full access to model parameters and gradients [14], whereas

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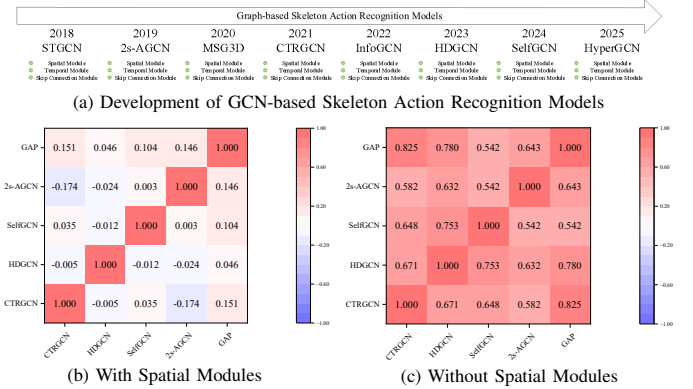


Fig. 1: (a) Structural evolution of representative GCN-based skeleton action recognition models from 2018 to the presents [1], [8], [9], [10], [11], [12], [2], [3], revealing a shared spatio-temporal decomposition into spatial, temporal, and skip-connection modules. (b) Inter-model gradient similarity with spatial modules enabled, showing substantial architectural divergence. (c) Inter-model gradient similarity with spatial modules removed for analysis, where increased similarity indicates that spatial modeling introduces model-specific gradients that limit transferability [13].

black-box attacks rely on extensive query interactions with the target system [15]. In contrast, transfer attacks generate adversarial samples on a surrogate model and deploy them against unseen target models without any access to parameters, internal states, or runtime interfaces [16]. Due to this model-agnostic nature, transfer attacks pose a broader and more practical threat, as a single adversarial sample may compromise multiple systems simultaneously [17], [18].

Following the transfer attack setting, existing analysis-oriented studies often adopt an algorithm-level perspective on skeleton-based action recognition to analyze the intrinsic transferability of adversarial perturbations across different GCN architectures. Despite recent progress, existing transfer attack methods, such as MIG [19], ResPA [20], and TASAR [21], still exhibit limited cross-model transferability and often incur noticeable perceptual distortion. Improvements in the transferability of these methods are often achieved at the cost of increased perceptual distortion, resulting in a suboptimal trade-off between attack effectiveness and imperceptibility. These limitations indicate that transferability is constrained by deeper architectural factors that remain insufficiently understood.

A key observation is that adversarial transferability is largely governed by architectural inductive biases shared across mod-

els. To investigate this issue, we conduct a systematic structural analysis of representative skeleton-based action recognition models. As shown in Figure 1(a), since the introduction of STGCN [1] in 2018, most models [8], [9], [10], [11], [12], [2], [3] have followed a spatio-temporal decomposition paradigm consisting of spatial modules, temporal modules, and skip connections modules. While recent studies [12], [2], [3] have primarily focused on enhancing spatial modeling through increasingly sophisticated skeleton graph structures, the temporal modules across different models remain highly similar in both structure and function. This asymmetry suggests that temporal modules tend to encode more universal motion patterns, whereas spatial modules introduce strong model-specific inductive biases.

To further understand the impact of these modules, we analyze inter-model gradient patterns. As illustrated in Figures 1(b) and 1(c), when spatial modules are included, the gradient similarity among different models is relatively low. In contrast, when spatial modules are disabled for analysis purposes, the gradient similarity among models increases significantly, indicating that spatial modeling is a major source of gradient misalignment. This observation implies that spatial modules, through their associated gradient dominance, substantially hinder the transferability of adversarial perturbations by amplifying model-specific characteristics.

Since temporal modules encode more universal motion patterns, shifting perturbations toward temporal components can better exploit shared structures across models and improve transferability. Motivated by this analysis, we hypothesize that attenuating the influence of spatial gradients, rather than eliminating spatial information entirely, can reduce model-specific bias and enhance cross-model transferability. Based on this insight, we propose a spatial gradient damper that explicitly suppresses dominant spatial gradients during adversarial optimization. By rebalancing gradient contributions, the perturbation budget is effectively shifted toward temporal components that are more consistent across different architectures.

However, improved transferability alone is insufficient for practical adversarial attacks if the resulting perturbations are perceptually implausible. Skeleton sequences represent articulated 3D structures, whereas human observers perceive them through 2D projections from limited viewpoints [22], [23]. As a result, perceptual sensitivity varies across viewpoints, and distortions are more likely to be noticed from commonly observed angles. To address this issue, we introduce a viewpoint-based constraint mechanism that enforces stricter constraints on common viewpoints while allowing greater flexibility on less frequent ones. This strategy further improves perceptual quality without compromising attack effectiveness.

By combining the spatial gradient damper with the viewpoint-based constraint mechanism, the proposed Spatial-Viewpoint Transfer Attack (SVAttack) simultaneously enhances transferability and perceptual imperceptibility. The proposed method shifts perturbations toward transferable temporal dynamics while preserving visually plausible skeleton motions. Extensive experiments on multiple benchmark datasets and target models demonstrate the superiority of SVAttack over existing transfer attack methods.

In summary, our main contributions are threefold. First, we identify, through structural and gradient-based analysis, that spatial modules introduce strong model-specific inductive biases that limit adversarial transferability in skeleton-based action recognition. Second, we propose a spatial gradient damper that suppresses dominant spatial gradients and effectively shifts perturbations toward more transferable temporal dynamics without increasing the perturbation budget. Third, we introduce a viewpoint-based perceptual constraint that exploits differences in human visual sensitivity across viewpoints, further enhancing imperceptibility while maintaining attack effectiveness. Extensive experimental results validate the effectiveness and robustness of the proposed SVAttack.

II. RELATED WORK

A. Skeleton-based Action Recognition

Most skeleton-based action recognition methods leverage graph convolutional networks to module spatial and temporal dependencies. STGCN [8] pioneered combining spatial GCNs with temporal convolutional networks for effective joint motion modeling. Subsequent methods, such as 2s-AGCN [1], MSG3D [9], and ASGCN [24], improved graph structures to better capture human skeletal dynamics. Recent models like InfoGCN [11], CTRGCN [10], GAP [25], SelfGCN [2], HDGCN [12], and HyperGCN [3] introduce adaptive channel and hierarchical spatial modeling to extract more discriminative features. These models adopt highly diverse spatial modeling strategies, resulting in substantial architectural heterogeneity, which poses fundamental challenges to the transferability of adversarial perturbations across different GCN-based action recognition models.

B. Adversarial Attacks on Skeleton-based Models

Adversarial attacks exploit model sensitivity by introducing small perturbations to input data to mislead predictions. Attacks are commonly categorized as white-box, black-box, or transfer attacks based on attacker access to model information. In white-box attacks, full access to model parameters and gradients allows highly effective perturbations [26], [15], [27], [28]. Representative methods include IFGSM [15], MIFGSM [27], and SMART [28], which can generate imperceptible perturbations but require full model knowledge. Black-box attacks, in contrast, rely solely on input-output queries [29], [30], trading attack effectiveness for practical applicability. For example, BASAR [31] uses a boundary search with inverse kinematics to generate skeleton adversarial examples but requires many queries to the target system. Transfer attacks have been proposed to overcome these limitations [19], [32], [20], [21]. These attacks generate adversarial examples on surrogate models, which are effective against unseen target models, eliminating the need for direct access. In skeleton action recognition models where spatial module differences introduce strong model-specific inductive biases. Addressing these architectural factors is crucial for designing transferable and perceptually plausible adversarial attacks.

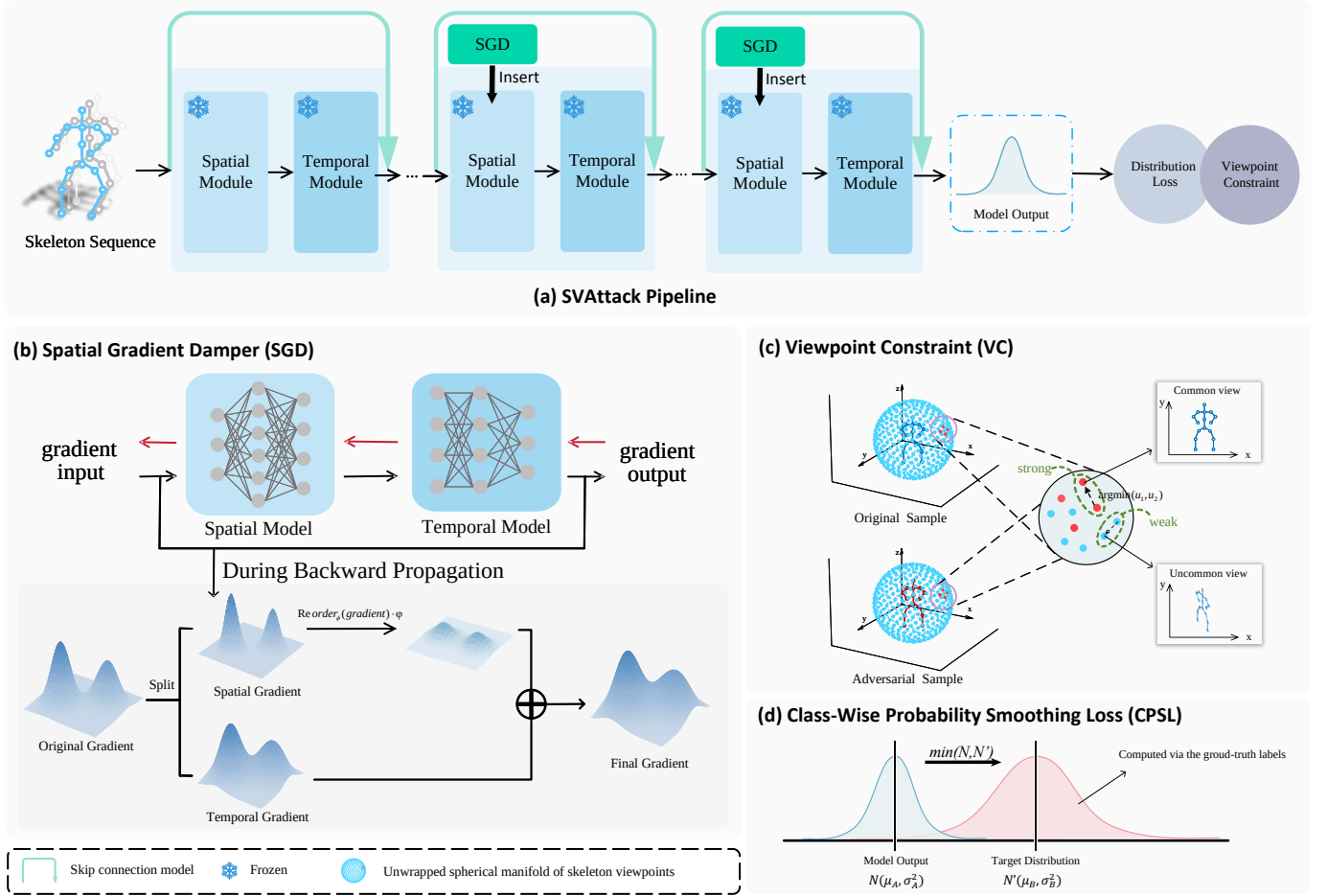


Fig. 2: Overall Framework of SVAttack. The framework consists of (a) a unified iterative attack pipeline, (b) the Spatial Gradient Damper for mitigating architecture-dependent bias, (c) the Viewpoint Constraint for perceptual plausibility, and (d) the Class-Wise Probability Smoothing Loss for enhancing cross-model transferability.

III. METHOD

Figure 2 provides an overview of the proposed SVAttack framework. Unlike most existing transfer-based skeleton attacks that primarily rely on surrogate-model decision boundary optimization, SVAttack explicitly addresses architectural and optimization factors that hinder cross-model transferability, while simultaneously accounting for perceptual plausibility under commonly used visualization settings.

To this end, we decompose the attack objective into three complementary components, each targeting a specific limitation observed in prior skeleton-based attacks. Specifically, the Spatial Gradient Damper (SGD) mitigates architecture-dependent spatial bias by suppressing dominant spatial gradients that tend to overfit surrogate models; the Viewpoint Constraint (VC) promotes perceptual stability under visually salient viewpoints, preventing structurally implausible perturbations; and the Class-Wise Probability Smoothing Loss (CPSL) discourages overconfident, target-specific optimization by encouraging decision-level confusion across action categories.

These components are jointly optimized within a unified iterative framework. In the following, we first describe the overall optimization framework and the interaction among

these components, and then detail the formulation and implementation of each module.

A. SVAttack Pipeline

As illustrated in Figure 2(a), SVAttack generates adversarial skeleton sequences through an iterative optimization process that jointly considers attack effectiveness and perceptual plausibility. Starting from a clean skeleton sequence, the adversarial sample is progressively updated by combining attack-driven gradients with perception-aware constraints. In particular, gradients from the proposed Class-Wise Probability Smoothing Loss and Viewpoint Constraint are explicitly incorporated, while spatial gradients are selectively modulated by the Spatial Gradient Damper (SGD) to mitigate model-specific bias.

By suppressing dominant and architecture-dependent spatial gradients and preserving perceptually salient viewpoints, SVAttack encourages perturbations to concentrate on transferable temporal dynamics while maintaining structural consistency in human-preferred views. This unified design enables SVAttack to achieve strong cross-model transferability without introducing noticeable perceptual artifacts.

Algorithm 1 details the iterative optimization procedure of SVAttack under the transfer attack setting. At each iteration,

Algorithm 1 SVAttack: Transferable Adversarial Attack for Skeleton-based Action Recognition

- 1: **Input:** surrogate model F ; Class-Wise Probability Smoothing Loss J ; Viewpoint Loss V ; 2D projection function $Flatten2d$; Spatial Gradient Damper SGD ; clean skeleton sequence x ; step size α ; trade-off weights γ, μ ;
- 2: **Initialize:** $\mathbf{x}_{adv} = x$
- 3: **Output:** adversarial skeleton sequence $\mathbf{x}_{adv}$
- 4: **for** $i = 0$ **to** $T - 1$ **do**
- 5: $output = SGD(F(\mathbf{x}_{adv}))$
- 6: Compute the Class-Wise Probability Smoothing Loss gradient $\nabla J(output)$
- 7: $v1 = \mu \nabla V(\mathbf{x}_{adv}, x)$
- 8: $v2 = \nabla V(Flatten2d(\mathbf{x}_{adv}), Flatten2d(x))$
- 9: Combine gradients:
- 10: $\nabla G = \gamma \nabla J + (1 - \gamma)(v1 + v2)$
- 11: Update adversarial sample:
- 12: $\mathbf{x}_{adv} = \mathbf{x}_{adv} + \alpha \cdot \nabla G$
- 13: **end for**
- 14: **return** $\mathbf{x}_{adv}$

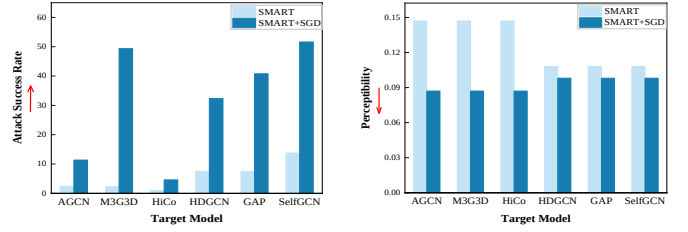
SVAttack computes the CPSL and obtains a damped gradient via SGD based on this loss, which is then combined with VC to limit the perceptual distortion caused by the adversarial perturbation. These gradients are combined through a weighted aggregation strategy to update the adversarial sample.

B. Spatial Gradient Damper (SGD)

Modern GCN-based skeleton action recognition models are typically composed of spatial modules, temporal modules, and skip connections modules, among which spatial modeling plays a dominant role in achieving high recognition accuracy. While effective for in-distribution performance, this strong reliance on spatial modeling amplifies architecture-specific inductive biases, leading to gradients that are highly coupled with surrogate-model decision boundaries and thus limiting cross-model adversarial transferability. In contrast, temporal dynamics encode more universal motion patterns that are shared across different model architectures, making temporal perturbations inherently more transferable.

From a gradient composition perspective, excessive spatial gradients tend to dominate the update direction during adversarial optimization, leaving limited capacity for perturbations along more transferable temporal dimensions. Motivated by this observation, we propose to moderately suppress spatial gradients to reduce model-specific bias and improve generalization in transfer attack settings. Importantly, we do not eliminate spatial information entirely, as indiscriminate suppression may hinder optimization stability and impede effective attack convergence.

Instead, we hypothesize that spatial gradients from later layers contribute more strongly to surrogate-specific decision boundaries than those from early layers. This is because later spatial layers encode higher-level joint interactions and semantic action patterns, whose gradients are more tightly



(a) Cross-model transferability (b) Perceptibility

Fig. 3: Illustrative analysis of the proposed Spatial Gradient Damper (SGD). This figure provides an intuitive motivation for suppressing dominant spatial gradients by showing its effect on cross-model transferability under a fixed perturbation budget. (a) Relative transfer performance of the SMART attack [28] with and without SGD across multiple target models. (b) Corresponding perceptual distortion trends, indicating that spatial gradient damping does not introduce noticeable visual artifacts. All results are obtained on NTU60 [33] using STGCN [1] and CTRGCN [10] as representative surrogate models.

coupled with model-dependent representations, whereas early layers primarily capture generic skeletal structures that are largely shared across architectures. Based on this insight, we design a Spatial Gradient Damper (SGD) that selectively attenuates dominant spatial gradients in the later layers of the spatial module.

As illustrated in Figure 2(b), SGD operates purely at the gradient level during backpropagation, without modifying the forward inference or network parameters. Specifically, spatial gradients are reweighted before being combined with other gradient components to form the final adversarial update direction. This design reduces architecture-dependent spatial bias while encouraging perturbations to shift toward more transferable temporal dynamics.

Formally, the damped spatial gradient contribution is defined as:

$$\mathbf{G}^e = \sum_{n=e}^L \varphi \cdot \text{Reorder}_{\phi}(\nabla \mathcal{L}^n), \quad (1)$$

where $\nabla \mathcal{L}^n$ denotes the gradient of the loss with respect to the spatial module at the n -th layer. $\text{Reorder}_{\phi}(\cdot)$ selects the top- ϕ fraction of spatial gradients according to their magnitude, ensuring that only the most dominant and potentially model-specific components are attenuated.

This selective design preserves early-layer structural cues while reducing high-level spatial bias that impairs transferability. To provide an intuitive justification for this design choice, we present an illustrative example by incorporating SGD into a representative skeleton-based attack, namely SMART [28]. As shown in Figure 3, suppressing dominant spatial gradients leads to more transferable perturbations, without introducing noticeable perceptual distortion.

C. Viewpoint Constraint (VC)

Existing skeleton-based adversarial attacks typically impose kinematic or smoothness constraints on the entire 3D skeleton

sequence. While these constraints help maintain motion coherence, they do not explicitly account for perceptual plausibility under common visualization conditions, where a 3D skeleton is ultimately observed as a 2D projection. Perturbations that distort the 2D projection can lead to visually implausible skeletons, even if the 3D kinematics remain feasible.

To address this, we introduce a Viewpoint Constraint (VC), which enforces stability in visually salient views while allowing flexibility in less critical projections. Formally, given a 3D skeleton sequence X and its adversarial counterpart X^{adv} , we define a differentiable 2D projection function $Flatten2d(\cdot)$ that maps each 3D frame into a corresponding 2D view:

$$X^{2d} = Flatten2d(X), \quad (2)$$

$$X^{adv,2d} = Flatten2d(X^{adv}). \quad (3)$$

The viewpoint-aware perception loss is then composed of three complementary terms: (i) joint position deviation, (ii) bone length deviation, and (iii) acceleration deviation. These terms are computed for both 2D projections and the original 3D skeleton:

$$\begin{aligned} \mathcal{L}_{2d} = & \beta_0 \|X^{2d} - X^{adv,2d}\|_2^2 \\ & + \beta_1 \frac{1}{M} \sum_{n=0}^{M-1} \|B_n^{2d} - B_n^{adv,2d}\|_2^2 \\ & + \beta_2 \|\ddot{X}^{2d} - \ddot{X}^{adv,2d}\|_2^2, \end{aligned} \quad (4)$$

$$\begin{aligned} \mathcal{L}_{3d} = & \beta_0 \|X - X^{adv}\|_2^2 \\ & + \beta_1 \frac{1}{M} \sum_{n=0}^{M-1} \|B_n - B_n^{adv}\|_2^2 \\ & + \beta_2 \|\ddot{X} - \ddot{X}^{adv}\|_2^2, \end{aligned} \quad (5)$$

$$\mathcal{L}_{perception} = \mathcal{L}_{2d} + \mu \mathcal{L}_{3d}. \quad (6)$$

Here, M is the total number of frames in the sequence, B denotes skeletal bone lengths, and $\ddot{X}$ represents the temporal acceleration of joint positions. The weights $\beta_0, \beta_1, \beta_2 \in [0, 1]$ control the relative importance of each term, and μ balances the contributions of 3D and 2D constraints.

By enforcing consistency in 2D projections while maintaining 3D kinematic constraints, the VC ensures that adversarial perturbations produce skeleton sequences that are both structurally coherent and perceptually plausible across common viewpoints, without relying on any subjective assumptions about human perception.

D. Class-Wise Probability Smoothing Loss (CPSL)

Existing adversarial attack methods for skeleton-based action recognition commonly generate perturbations by directly reducing the confidence of the ground-truth class through the cross-entropy loss [34]. While effective in white-box settings, such objectives often produce highly surrogate-specific perturbations, which limits their transferability across different model architectures.

To mitigate this issue, we propose a *Class-Wise Probability Smoothing Loss* (CPSL), which aims to reduce excessive class-wise probability concentration and encourage a balanced redistribution of prediction confidence across classes. Here, the term “smoothing” refers to alleviating dominant class confidence rather than enforcing a uniform or entropy-maximizing

distribution. Figure 2(d) illustrates the integration of CPSL within the SVAttack framework.

Formally, let C denote the total number of action classes, and let x_i be the i -th input sample with predicted probability vector $p_i \in \mathbb{R}^C$. We define a class-wise balanced target distribution y_i^* as

$$y_{i,j}^* = \begin{cases} \rho, & j = y_{\text{true}}, \\ \frac{1-\rho}{C-1}, & j \neq y_{\text{true}}, \end{cases} \quad (7)$$

where y_{true} denotes the ground-truth class label of x_i . This formulation preserves a moderate bias toward the true class while preventing overly confident, peaked predictions that are tightly coupled to surrogate-specific decision boundaries.

Rather than matching the target distribution via cross-entropy or Kullback–Leibler divergence, CPSL aligns the first- and second-order statistics of the predicted and target distributions. For a batch of N samples, the loss terms are defined as

$$\mathcal{L}_\mu = \frac{1}{N} \sum_{i=1}^N (\mathbb{E}[y_i^*] - \mathbb{E}[p_i])^2, \quad (8)$$

$$\mathcal{L}_\sigma = \frac{1}{N} \sum_{i=1}^N (\text{Var}(y_i^*) - \text{Var}(p_i))^2, \quad (9)$$

$$\mathcal{L}_{\text{CPSL}} = \mathcal{L}_\mu + \mathcal{L}_\sigma, \quad (10)$$

where $\mathbb{E}[\cdot]$ and $\text{Var}[\cdot]$ denote the mean and variance computed over class probabilities for each sample.

Matching the mean encourages global redistribution of probability mass across classes, while matching the variance constrains the sharpness of the predicted distribution. Together, these two objectives prevent over-confident predictions that are strongly tied to surrogate-specific representations. As a result, CPSL guides adversarial optimization toward probability dispersion patterns that are less sensitive to architectural differences.

IV. EXPERIMENTS

A. Experimental Setup

In this section, we describe the experimental setup used to evaluate the effectiveness and transferability of the proposed SVAttack method. We detail the datasets, target models, baseline attacks, proxy models for transfer attacks, defense methods, and implementation details to ensure reproducibility and fair comparison. Unless otherwise stated, all perturbations are constrained by a maximum budget of $\epsilon = 0.01$ under the L_∞ norm. In the spatial gradient damping module, the damping factor φ is set to 0.1, and the starting layer index e is set to 4. In the Viewpoint Constraint module, the hyperparameters μ , β_0 , β_1 , and β_2 are set to 0.2, 0.6, 0.3, and 0.4, respectively. In the Class-Wise Probability Smoothing Loss, the hyperparameter ρ is set to 0. In Algorithm 1, the hyperparameters α and γ are set to 0.01 and 0.6, respectively. All experiments are conducted on NVIDIA A100-SXM4-80GB GPUs.

Dataset. We evaluate our method on three publicly available skeleton-based action recognition datasets: NTU60 [33], NTU120 [35], and Kinetics [36]. The NTU60 dataset contains

60 activity classes and 56,880 skeleton sequences, covering a wide range of daily human actions. The NTU120 dataset extends NTU60 to 120 activity classes with 113,945 sequences, providing a more comprehensive set of human activities. The skeletons in NTU60 and NTU120 are represented in a 25-joint format and normalized for consistency. The Kinetics dataset is derived from the Kinetics-400 video dataset using OpenPose for pose estimation. It contains 240,436 training sequences and 19,796 testing sequences across 400 action categories. Each frame provides 18 body joints, and for clips with multiple people, the two individuals with the highest average joint confidence are selected.

Target Model. We consider nine state-of-the-art models as attack targets: STGCN [8], 2s-AGCN [1], MSG3D [9], HiCo [37], CTRGCN [10], GAP [25], SelfGCN [2], HDGCN [12], and HyperGCN [3]. All models are trained using their official implementations and recommended training protocols. For 2s-AGCN, we specifically attack the single joint flow variant (js-AGCN). To evaluate transferability against heterogeneous architectures, we include the HiCo model. HiCo is an unsupervised model based on a Transformer architecture, in contrast to other models that rely on GCNs. This unsupervised heterogeneous setting poses a greater challenge for the proposed attack.

Baselines. We compare our method with seven representative baselines: I-FGSM [15], MI-FGSM [27], SMART [28], MIG [19], SVRE [32], ResPA [20], and TASAR [21]. All baselines are evaluated under the same perturbation budget to ensure fair comparison. We select two representative GCN architectures, STGCN and CTRGCN, as proxy models for transfer attacks. STGCN serves as the proxy for MSG3D, 2s-AGCN, HiCo, and similar target models. CTRGCN is used as the proxy for HDGCN, GAP, SelfGCN, and HyperGCN. This selection reflects differences in input requirements: STGCN and its targets process 300 frames per sequence, while CTRGCN and its targets process 64 frames. By distinguishing proxy models, we simulate transfer attacks under heterogeneous input scenarios. This design allows a rigorous evaluation of transferability across models and ensures broad applicability of the results.

Defense Methods. We evaluate our attack against two representative defense strategies: BEAT [38] and TRADES [39]. BEAT constrains the energy of adversarial perturbations through sequentially appended auxiliary modules. We appended five auxiliary modules to the base model and trained each module for 10 epochs. This layer-wise training enhances the model's robustness and stabilizes predictions under adversarial attacks. TRADES enforces a trade-off between natural accuracy and adversarial robustness via regularized adversarial training. We configure TRADES with 50 training epochs, a learning rate of 0.01, an ϵ of 0.01, $\beta = 1.0$, and the Linf norm for perturbation measurement. These settings allow TRADES to improve robustness while maintaining reasonable accuracy on clean data.

B. Evaluation Metrics

Perceptibility Metrics. We adopt kinematic metrics to evaluate the perceptibility of adversarial skeleton sequences [40],

which quantify deviations in joint coordinates, bone lengths, and temporal accelerations. Smaller values indicate higher similarity to the original skeletons and better perceptual plausibility.

Formally, given a batch of M sequences each with T frames and L joints, the overall kinematic deviation is defined as:

$$\begin{aligned} \Delta P = & \frac{1}{MTL} \sum_{i=1}^M \sum_{t=1}^T \|X_{i,t} - X_{i,t}^{adv}\|_2 \\ & + \frac{1}{MTL} \sum_{i=1}^M \sum_{t=1}^T \|B_{i,t} - B_{i,t}^{adv}\|_2 \\ & + \frac{1}{MTL} \sum_{i=1}^M \sum_{t=1}^T \|\ddot{X}_{i,t} - \ddot{X}_{i,t}^{adv}\|_2, \end{aligned} \quad (11)$$

where X , B , and $\ddot{X}$ denote joint coordinates, bone lengths, and temporal accelerations, respectively.

Perceptibility metrics are computed in two representation spaces:

- ΔP_{3D} measures physical plausibility and motion consistency directly in 3D skeleton space.
- ΔP_{2D} evaluates visual consistency after projecting skeleton sequences onto image planes. For ΔP_{2D} , skeletons are projected using a fixed elevation angle of 45° and an azimuth angle of 35° (see Section III-C).

Both metrics share the same mathematical formulation; they differ only in the space (3D or 2D) where the computation is performed.

Attack Success Rate (ASR). The attack success rate (ASR) measures the effectiveness of adversarial attacks in misleading the target model. We consider two types of ASR:

- Untargeted attack ASR: the proportion of adversarial samples that cause the model to predict any class other than the true class.
- Targeted attack ASR: the proportion of adversarial samples that successfully force the model to predict a specific target class.

Formally, ASR is defined as:

$$\text{ASR} = \frac{N_{\text{succ}}}{N_{\text{total}}} \times 100\%, \quad (12)$$

where N_{succ} denotes the number of successfully attacked samples (depending on the attack type) and N_{total} is the total number of adversarial samples. A higher ASR indicates a more effective attack. Unless otherwise specified, the reported ASR refers to the untargeted attack type.

C. Comparison of Attack Results

Untargeted Attack Analysis. In Tables I and II, we report the untargeted attack results using STGCN and CTRGCN as proxy models on the NTU-60, NTU-120, and Kinetics datasets. The results show that SVAttack attains higher attack success rates while maintaining lower perceptibility metrics, indicating that the improved transferability is not obtained at the cost of excessive perturbations.

It is worth noting that HiCo adopts a Transformer-based unsupervised learning framework, which differs substantially from the GCN-based proxy models used for attack generation. Such architectural and optimization discrepancies make

TABLE I: Attack success rates (ASR, %) against different target models under STGCN and CTRGCN proxy settings on NTU60 and NTU120. ΔP_{2D} denotes the perceptibility metric (lower is better). Avg represents the mean ASR across target models under the same proxy model.

Dataset	Method	STGCN as Proxy Model					CTRGCN as Proxy Model					
		Percep.	Target Models ASR (%)				Percep.	Target Models ASR (%)				
		$\Delta P_{2D} \downarrow$	2s-AGCN	MSG3D	HiCo	Avg $\uparrow$	$\Delta P_{2D} \downarrow$	HDGCN	SelfGCN	GAP	Hyper-GCN	Avg $\uparrow$
NTU60	I-FGSM	0.518	5.30%	7.63%	4.20%	5.71%	0.681	3.30%	4.60%	1.20%	27.59%	9.68%
	MI-FGSM	0.697	8.10%	12.50%	1.61%	7.40%	0.910	7.50%	7.40%	3.50%	41.98%	15.60%
	SVRE	0.987	9.09%	11.18%	2.76%	7.01%	1.488	25.78%	26.60%	30.82%	43.03%	31.56%
	SMART	0.147	2.39%	2.25%	1.00%	1.88%	0.108	7.50%	7.50%	13.75%	31.66%	15.10%
	MIG	0.416	5.48%	5.48%	1.37%	4.11%	1.146	16.15%	19.23%	20.38%	29.61%	21.34%
	ResPA	0.505	3.21%	6.81%	1.00%	3.67%	1.492	10.80%	13.43%	15.86%	19.71%	14.45%
	TASAR	1.598	7.13%	15.72%	2.81%	8.55%	1.796	14.50%	14.57%	26.26%	35.21%	22.14%
	SVAttack	0.085	12.95%	51.95%	4.20%	23.03%	0.096	34.37%	42.34%	52.94%	52.04%	45.42%
NTU120	I-FGSM	0.529	3.30%	4.60%	1.20%	3.70%	0.718	28.40%	32.86%	23.51%	22.99%	26.44%
	MI-FGSM	0.716	7.50%	7.40%	3.50%	6.13%	0.907	31.53%	37.81%	28.87%	28.43%	31.66%
	SVRE	1.072	9.53%	9.53%	7.17%	8.08%	1.479	32.38%	32.28%	34.47%	40.77%	34.98%
	SMART	0.133	1.60%	7.70%	1.20%	3.50%	0.159	5.74%	8.10%	10.44%	11.69%	8.49%
	MIG	0.503	6.63%	7.69%	4.46%	6.26%	0.983	25.53%	30.93%	22.45%	28.63%	26.39%
	ResPA	0.333	0.60%	1.60%	0.20%	0.80%	0.505	10.56%	10.97%	15.58%	16.73%	13.21%
	TASAR	1.697	7.87%	13.21%	3.43%	8.17%	1.620	21.71%	27.81%	26.97%	40.15%	29.66%
	SVAttack	0.109	33.50%	59.21%	5.90%	32.87%	0.111	33.30%	42.07%	43.52%	41.23%	40.03%

transfer-based attacks particularly challenging. Nevertheless, as shown in Table I, SVAttack maintains competitive performance on both NTU-60 and NTU-120, suggesting enhanced cross-architecture transferability.

Overall, the results in Tables I and II provide strong empirical evidence that suppressing the spatial module contributes to improved transferability in skeleton-based adversarial attacks.

In Table III, we further report the 3D and 2D perceptibility metrics evaluated on STGCN₁₂₀, CTRGCN₆₀, and CTRGCN₁₂₀. Compared with other methods, SVAttack achieves lower values on ΔC , ΔB and ΔA , indicating that the proposed approach does not obtain transferability at the cost of perceptibility. Moreover, it can be observed that the values on ΔA are higher than those on ΔC and ΔB , which effectively verifies that the perturbations are mainly concentrated in the temporal dimension.

Targeted Attack Analysis. Table IV presents the targeted attack results using STGCN and CTRGCN as proxy models. As expected, all methods exhibit a noticeable decrease in success rate compared to the untargeted setting, reflecting the increased difficulty of targeted transfer-based attacks. In this scenario, the adversarial samples must not only induce misclassification on the proxy model but also consistently transfer the same target label to different victim models. Discrepancies in network architectures, training objectives, and optimization dynamics between proxy and target models further exacerbate this challenge.

Despite these difficulties, SVAttack demonstrates superior performance in terms of both attack success rate and percepti-

bility. This suggests that the proposed spatial suppression and viewpoint constraint mechanisms help stabilize perturbation patterns, which is beneficial even under stricter targeted attack settings.

Defense Analysis. As shown in Table V, we generate adversarial samples using STGCN as the proxy model and evaluate their transferability under BEAT and TRADES defense mechanisms on 2s-AGCN and MSG3D. This experimental setting enables an analysis of how different defense strategies influence attack effectiveness and cross-model transferability. Compared with baseline methods, SVAttack maintains relatively strong attack effectiveness under both defense mechanisms. These results suggest that the proposed method exhibits improved resilience to defense strategies, further highlighting its transferability across diverse model configurations.

D. Ablation Experiment

Ablation Analysis of Modules. The results in Table VI show that removing either the spatial gradient damper (SGD) or the Class-Wise Probability Smoothing Loss (CPSL) leads to a degradation in performance, validating the effectiveness of each proposed component. In particular, removing SGD causes a substantial drop in attack success rate, which supports our hypothesis that suppressing spatial gradients is critical for enhancing transferability.

Figures 4 and 5 illustrate perceptibility evaluations across different viewpoints. Without Viewpoint Constraints (VC), perceptibility increases noticeably near visually salient viewpoints, such as when the azimuth angle approaches 45° or

TABLE II: Attack success rates (ASR, %) against different target models under STGCN and CTRGCN proxy settings on the Kinetics dataset. ΔP_{2D} denotes the perceptibility metric (lower is better). Avg represents the mean ASR across target models under the same proxy model.

Dataset	Method	STGCN as Proxy Model					CTRGCN as Proxy Model				
		Percep.	Target Models ASR (%)				Percep.	Target Models ASR (%)			
		$\Delta P_{2D} \downarrow$	2s-AGCN	MSG3D	HiCo	Avg $\uparrow$	$\Delta P_{2D} \downarrow$	HDGCN	SelfGCN	Hyper-GCN	Avg $\uparrow$
Kinetics	I-FGSM	0.452	2.50%	1.78%	8.75%	4.34%	1.076	28.76%	20.43%	33.84%	27.01%
	MI-FGSM	0.634	5.00%	3.00%	11.80%	6.60%	1.213	31.55%	23.71%	31.77%	29.68%
	SVRE	0.897	6.94%	7.29%	18.40%	10.88%	1.027	19.03%	12.50%	24.89%	18.81%
	SMART	0.233	6.87%	7.50%	14.37%	9.58%	0.262	32.89%	32.10%	31.75%	32.91%
	MIG	0.952	23.79%	15.08%	9.59%	16.82%	1.278	34.91%	27.81%	41.83%	34.85%
	ResPA	0.728	8.65%	5.85%	13.95%	9.48%	0.920	17.75%	12.95%	18.03%	16.91%
	TASAR	2.021	10.32%	13.28%	17.56%	13.05%	2.088	6.25%	17.97%	18.00%	14.74%
	SVAttack	0.146	31.04%	20.43%	10.21%	20.56%	0.173	36.54%	30.71%	52.26%	39.84%

TABLE III: Perceptibility Evaluation. STGCN₁₂₀ denotes adversarial samples generated by attacking the STGCN model on NTU-120, while CTRGCN_{60/120} represents adversarial samples generated by attacking the CTRGCN model on NTU-60/120.

Method	STGCN ₁₂₀		CTRGCN ₆₀		CTRGCN ₁₂₀	
	$\Delta P_{3D} \downarrow$	$\Delta P_{2D} \downarrow$	$\Delta P_{3D} \downarrow$	$\Delta P_{2D} \downarrow$	$\Delta P_{3D} \downarrow$	$\Delta P_{2D} \downarrow$
I-FGSM	4.143	0.529	5.528	0.681	5.737	0.718
MI-FGSM	5.591	0.716	7.385	0.910	7.264	0.907
SMART	1.036	0.133	0.804	0.108	0.928	0.159
ResPA	2.629	0.333	3.904	1.492	3.969	0.505
TASAR	13.382	1.697	14.467	1.796	14.563	1.820
SVAttack	0.891	0.109	0.701	0.096	0.817	0.111

the elevation angle approaches 35°. After introducing VC, perceptibility at these viewpoints is significantly reduced, demonstrating the effectiveness of the viewpoint constraint strategy.

Table VII reports the joint coordinate shift ΔC , bone length shift ΔB , and acceleration shift ΔA . The first two metrics reflect spatial perturbations, while the acceleration shift captures temporal variations. Lower values indicate better perceptibility. Methods such as ResPA and TASAR introduce relatively large perturbations in both spatial and temporal dimensions, but these perturbations do not translate into proportional gains in attack success rate. Although SMART generates adversarial samples with smaller ΔA on CTRGCN compared to SVAttack, its ΔC and ΔB shifts are noticeably higher than those of SVAttack, even exceeding its own ΔA , indicating that its perturbations fail to effectively focus on vulnerabilities shared across models. In addition, SMART's Δp_{3d} value is 14.55% higher than that of SVAttack. In contrast, SVAttack exhibit smaller spatial shifts while concentrating perturbations in the temporal dimension, which is consistent with their improved attack efficiency. This explains why SVAttack can achieve higher attack success rates under a smaller perturbation budget.

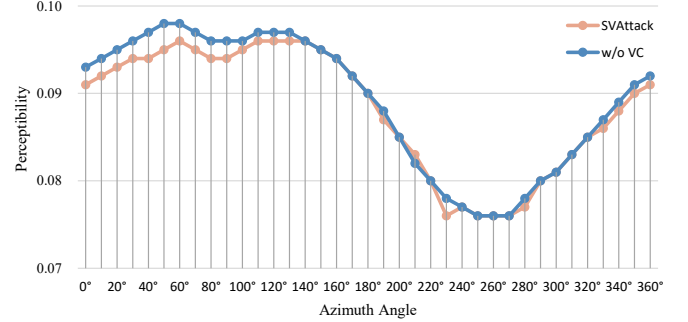


Fig. 4: Perceptual evaluation of SVAttack using samples generated by the CTRGCN proxy model at a fixed azimuth angle of 35° across different elevation angles. The blue line represents SVAttack without viewpoint constraints, and the orange line represents SVAttack with viewpoint constraints.

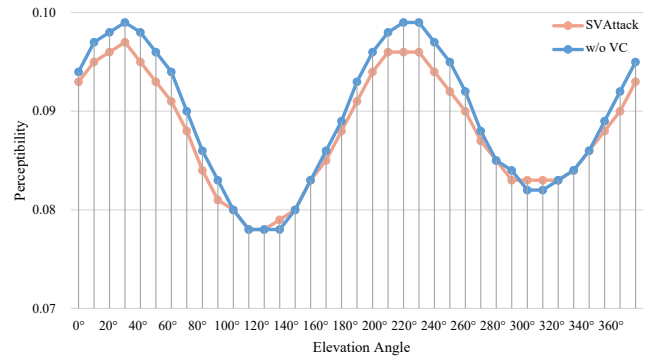


Fig. 5: Perceptual evaluation of SVAttack using samples generated by the CTRGCN proxy model at a fixed elevation angle of 45° across different azimuth angles. The blue line represents SVAttack without viewpoint constraints, and the orange line represents SVAttack with viewpoint constraints.

Adversarial Sample Category Distribution. Figure 6 presents the distribution of predicted categories on 2s-AGCN,

TABLE IV: Targeted attack success rates (ASR, %) against different target models under STGCN and CTRGCN proxy settings on NTU60. ΔP_{2D} denotes the perceptibility metric (lower is better). Avg represents the mean ASR across target models under the same proxy model.

Dataset	Method	STGCN as Proxy Model					CTRGCN as Proxy Model				
		Percep.	Target Models ASR (%)				Percep.	Target Models ASR (%)			
		$\Delta P_{2D} \downarrow$	2s-AGCN	MSG3D	HiCo	Avg $\uparrow$	$\Delta P_{2D} \downarrow$	HDGCN	GAP	SelfGCN	Avg $\uparrow$
NTU60	IFGSM	0.479	1.90%	1.70%	2.20%	1.93%	0.572	1.60%	1.60%	1.40%	1.15%
	MIFGSM	0.627	2.10%	2.30%	2.30%	2.23%	0.733	2.30%	3.60%	1.70%	2.53%
	SMART	0.213	2.20%	2.51%	2.90%	2.54%	0.135	3.20%	4.00%	3.60%	3.60%
	SVRE	1.092	3.00%	3.20%	2.60%	7.07%	1.588	2.60%	2.20%	2.80%	2.53%
	MIG	1.062	3.33%	3.00%	2.67%	3.00%	1.632	4.20%	3.20%	3.80%	3.73%
	TASAR	1.703	2.83%	3.16%	1.83%	2.61%	1.774	1.00%	1.00%	1.50%	1.17%
	SVAttack	0.900	7.98%	10.06%	10.52%	9.52%	0.126	4.33%	4.33%	3.20%	3.95%

TABLE V: Attack success rate (ASR, %) under BEAT and TRADE defenses on NTU60. The proxy model is the STGCN.

Defense	Target Model	I-FGSM	SMART	TASAR	SVAttack
BEAT	2s-AGCN	0.50%	1.90%	1.56%	14.70%
	MSG3D	7.13%	4.71%	12.96%	53.76%
TRADE	2s-AGCN	0.60%	1.41%	0.87%	4.16%
	MSG3D	1.23%	1.67%	2.12%	4.72%

TABLE VI: Ablation study of the SGD and CPSL modules on NTU-60. STGCN is used as the proxy model. ΔP_{2D} indicates perceptibility (lower is better), and Avg denotes the mean attack success rate across target models.

Method	Percep.	Target Models			Avg $\uparrow$
	$\Delta P_{2D} \downarrow$	2s-AGCN	MSG3D	HiCo	
w/o SGD	0.143	1.60%	5.22%	2.04%	8.86%
w/o CPSL	0.085	12.49%	49.39%	3.18%	21.68%
SVAttack	0.085	12.95%	51.95%	4.20%	23.03%

MSG3D, and HiCo when evaluated on adversarial samples generated by the STGCN proxy model. The predicted category distributions remain largely consistent across different models, indicating that SVAttack captures shared vulnerabilities among

TABLE VII: Detailed perception results of adversarial samples on the NTU60 dataset using ΔP_{3D} : ΔC denotes joint coordinate shift, ΔB denotes bone length shift, ΔA denotes acceleration shift and PM denotes the proxy model.

PM	Method	$\Delta C \downarrow$	$\Delta B \downarrow$	$\Delta A \downarrow$	PM	Method	$\Delta C \downarrow$	$\Delta B \downarrow$	$\Delta A \downarrow$
STGCN	I-FGSM	0.486	0.579	3.188	CTRGCN	I-FGSM	0.371	0.463	4.693
	MI-FGSM	0.665	0.797	4.257		MI-FGSM	0.511	0.639	6.234
	SMART	0.312	0.446	0.371		SMART	0.248	0.346	0.209
	ResPA	0.455	0.525	3.136		ResPA	0.264	0.311	3.329
	TASAR	1.022	1.185	11.268		TASAR	1.009	1.181	12.278
	SVAttack	0.133	0.210	0.307		SVAttack	0.100	0.146	0.455

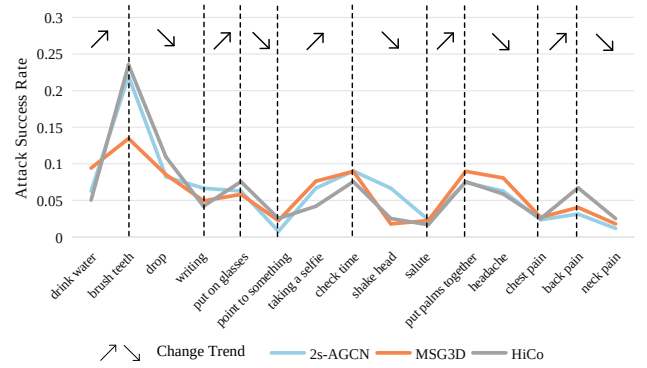


Fig. 6: Trends of attack success rate across different classes on three target models for adversarial samples generated by the STGCN proxy model.

diverse architectures and exhibits strong generalization capability.

Hyperparameter Ablation Analysis. Figure 7 shows the effect of the hyperparameter u in Equation (6) on the NTU-60 dataset. When $u = 0.2$, the attack success rate reaches its peak while perceptibility remains near-optimal. Balancing effectiveness and perceptibility, we adopt $u = 0.2$ in all subsequent experiments.

We further analyze the impact of additional hyperparameters in the extended formulation of Equation (10), denoted as Equation (13), as shown in Figure 8:

$$\mathcal{L}_{dif} = \alpha \mathcal{L}_{\mu} + \beta \mathcal{L}_{\sigma}. \quad (13)$$

Increasing α improves attack success rates but also increases perceptibility. Considering the trade-off between attack effectiveness and stability, as well as avoiding unnecessary hyperparameter tuning, we set $\alpha = \beta = 1$ in the final configuration.

Layer-Level Ablation Study. Figure 9 reports the layer-level ablation results on NTU-60 using CTRGCN as the surrogate model. The evaluation is conducted across three target models, including GAP, HDGCN, and SelfGCN. For both settings that select 3 or 7 spatial layers for gradient

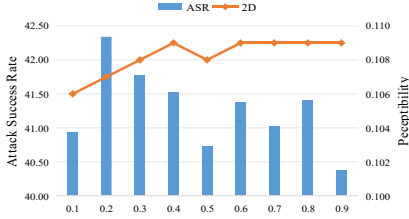


Fig. 7: Ablation of parameter μ in Eq. (6) on NTU-60 using CTRGCN as proxy to attack GAP.

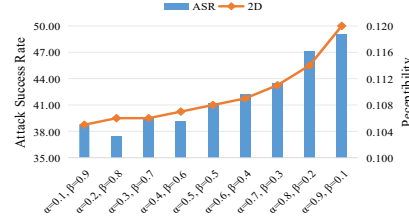


Fig. 8: Ablation of parameters α and β in Eq. (9) on NTU-60 using CTRGCN as proxy to attack GAP.

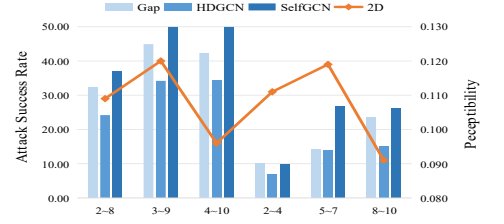


Fig. 9: Layer sensitivity ablation on NTU-60 using CTRGCN as proxy to attack GAP, HDGCN, and SelfGCN.

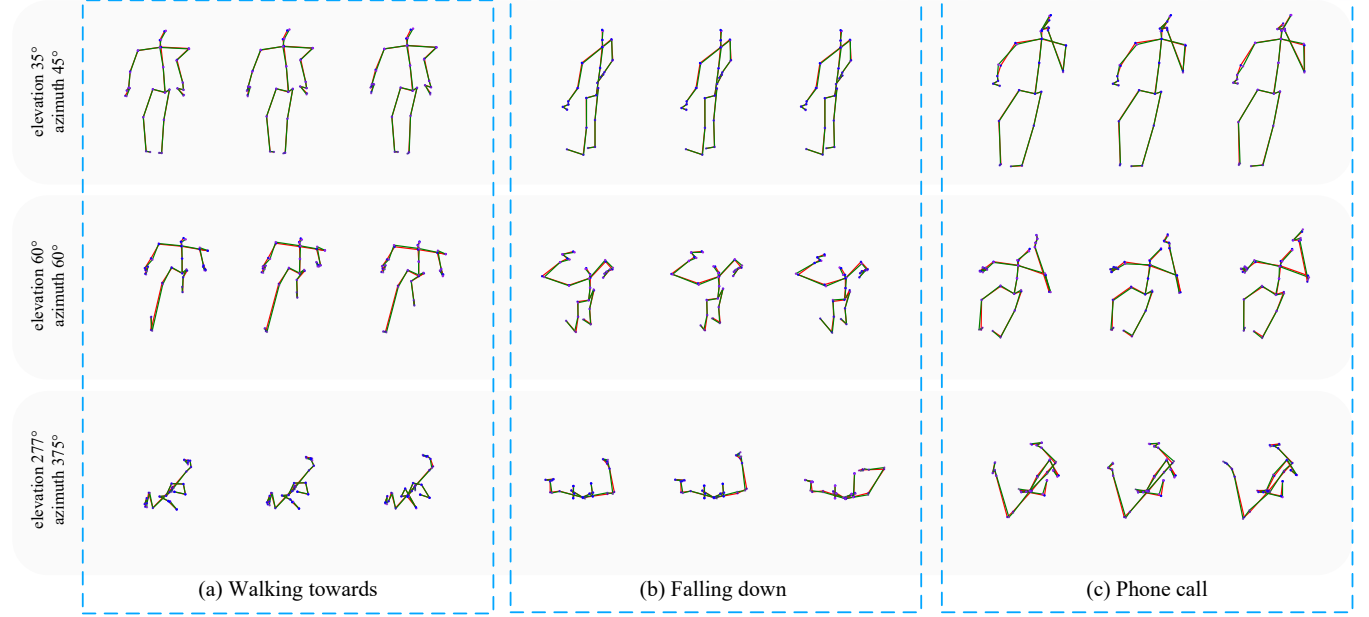


Fig. 10: Visualization of adversarial skeleton samples generated by SVAttack under different viewing angles. Rows correspond to different elevation–azimuth configurations: $(35^\circ, 45^\circ)$, $(60^\circ, 60^\circ)$, and $(277^\circ, 275^\circ)$. Columns represent three action classes, namely *Walking towards*, *Falling down*, and *Phone call*. For the same sample, variations in viewing angles lead to noticeable differences in human perceptual appearance. Green and red skeletons denote the original and adversarial samples, respectively.

manipulation, layers closer to the output tend to yield higher attack success rates while maintaining lower perceptibility. This trend is observed consistently across different target architectures, indicating that late-stage spatial representations contribute more strongly to transferable adversarial effects. These results suggest that focusing perturbations on deeper layers is beneficial for improving attack transferability while preserving perceptual quality.

E. Visualization

Visualization of Adversarial Samples from Different Viewpoints. Although adversarial perturbations are generated in the 3D skeleton space, their perceptual visibility to human observers is inherently viewpoint dependent. When a 3D skeleton sequence is projected onto a 2D plane, occlusion, joint overlap, and visibility variations may occur, potentially obscuring local structural changes introduced by adversarial perturbations. Consequently, the perceptibility of the same adversarial sample can vary substantially across different viewing

angles, motivating a systematic visualization analysis from multiple viewpoints.

Figure 10 visualizes adversarial samples under several representative viewpoints. Under certain viewing angles, severe joint overlap or occlusion prevents clear observation of perturbation-induced deviations, whereas viewpoints around 35° elevation and 45° azimuth provide a more complete and stable depiction of the skeletal structure. Such viewpoints enable clearer visual inspection of both spatial configuration and motion patterns, resulting in more consistent perceptual assessments. This observation aligns with the design rationale of the proposed Visual Viewpoint Constraints, which aim to ensure reliable perceptibility evaluation across samples.

Visualization under Different Perceptibility Metrics. Figure 11 presents adversarial samples with progressively increasing perceptibility scores. As the perceptibility value increases, the visual discrepancy between adversarial and original skeletons becomes more evident, with perturbation-induced structural and motion changes appearing increasingly distinguishable. This qualitative trend is consistent with the corresponding

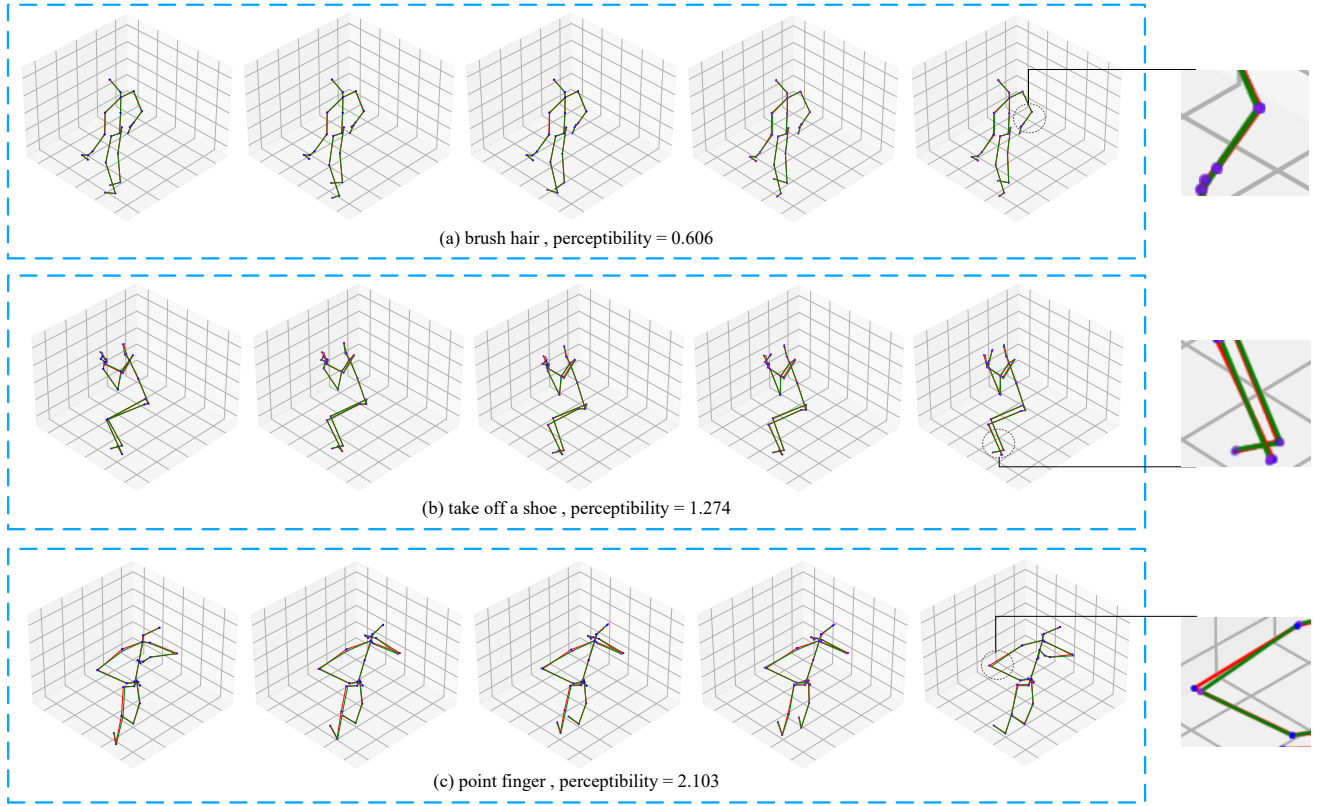


Fig. 11: Visualization of adversarial skeleton samples under different perceptibility levels. Green and red skeletons represent the original and adversarial samples, respectively. As perceptibility increases, adversarial perturbations become progressively more noticeable.

quantitative perceptibility measurements, providing empirical support for the effectiveness of the adopted kinematic-based perceptibility metrics.

V. CONCLUSIONS

This paper investigates the challenge of designing transferable and perceptually plausible adversarial attacks for skeleton-based action recognition models. We study transferability from an architectural and representational perspective. By conducting a systematic structural and gradient-based analysis, we reveal that spatial modeling introduces strong model-specific inductive biases that fundamentally limit cross-model transferability, whereas temporal modeling captures more universal motion patterns shared across architectures. Motivated by this insight, we propose to rebalance adversarial perturbations toward transferable temporal components while attenuating the influence of spatial gradients. Building on this analysis, we develop SVAttack, which integrates a spatial gradient damping strategy with a viewpoint-aware perceptual constraint. This combination enables the generation of adversarial samples that are both highly transferable across different GCN architectures and visually plausible under common viewpoints. Extensive experiments on multiple benchmark datasets demonstrate consistent improvements over existing transfer attack methods.

This work provides new insights into the architectural factors that govern adversarial transferability in skeleton-based

action recognition. Beyond the specific method presented here, our findings suggest a principled direction for designing transferable attacks and defenses by explicitly considering model inductive biases. Future research may extend these insights to end-to-end attack pipelines and broader multimodal perception systems.

ACKNOWLEDGMENTS

We acknowledge support from the National Natural Science Foundation of China (Grant No. 62406095), the Natural Science Foundation of Anhui Province (Grant No. 2308085MF213), as well as the Projects of Excellent Research and Innovation Teams in Anhui Province's Universities (Grant Nos. 2022AH010095 and 2024AH010030).

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